

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12419010
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Shearman; Simon J. et al.

Dual flow low profile coolant distribution manifold

Abstract

A coolant distribution manifold assembly for use in a module or circuit pack of an optical networking system, including: a body defining a main inlet port at one end, a main outlet port at another end, and a plurality of cooling plate inlet ports and cooling plate outlet ports disposed between the main inlet port and the main outlet port; where the body further defines an upper internal plenum and a lower internal plenum each coupled to one of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports. Optionally, the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port.

Inventors: Shearman; Simon J. (Carleton Place, CA), Mohajer; Behzad (Vancouver, CA), Ajersch; Peter (Ottawa, CA)

Applicant: Ciena Corporation (Hanover, MD)

Family ID: 1000008822366

Assignee: Ciena Corporation (Hanover, MD)

Appl. No.: 18/381265

Filed: October 18, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250133691 A1	Apr. 24, 2025

Publication Classification

Int. Cl.: H05K7/20 (20060101); H01R12/72 (20110101)

U.S. Cl.:

CPC **H05K7/20254** (20130101); **H01R12/725** (20130101); **H05K7/20281** (20130101);
H01R2201/04 (20130101)

Field of Classification Search

CPC: H05K (7/20254); H05K (7/20281); H01R (12/725); H01R (2201/04)

USPC: 361/688

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6807056	12/2003	Kondo	361/689	H05K 7/20781
7187549	12/2006	Teneketges	361/689	H01L 23/473
7336487	12/2007	Chrysler	361/679.48	H01L 23/473
7639499	12/2008	Campbell	165/104.19	H05K 7/20772
7710723	12/2009	Korich	174/15.1	H05K 7/20927
8279601	12/2011	Lima	361/695	G06F 1/20
8654532	12/2013	Chen	361/689	H05K 7/20781
9634959	12/2016	Coteus	N/A	H05K 7/20309
10749288	12/2019	Weltsch	N/A	B23P 19/04
11659696	12/2022	McEathron	361/679.47	H05K 7/20881
11980009	12/2023	Mohajer	N/A	H05K 7/20272
12262463	12/2024	Bubon	N/A	G01T 1/2985
2006/0002086	12/2005	Teneketges	257/E23.098	H01L 23/473
2006/0096738	12/2005	Kang	257/E23.098	F28F 3/022
2007/0062673	12/2006	Olesen	257/E23.098	F28F 9/0268
2007/0091570	12/2006	Campbell	361/699	H05K 7/2079
2009/0284921	12/2008	Colgan	361/699	H01L 23/473
2011/0182027	12/2010	Lima	361/679.46	G06F 1/20
2013/0105116	12/2012	Campbell et al.	N/A	N/A
2013/0333778	12/2012	Ferrarese	137/561A	F02C 7/228
2020/0006884	12/2019	Weltsch	N/A	H01R 13/514
2020/0229321	12/2019	Dogruoz et al.	N/A	N/A
2021/0112683	12/2020	Mohajer et al.	N/A	N/A
2022/0142013	12/2021	Zhou et al.	N/A	N/A
2023/0039781	12/2022	Mohajer et al.	N/A	N/A

OTHER PUBLICATIONS

Jan. 13, 2025 International Search Report issued in corresponding PCT Application No. PCT/US2024/051220. cited by applicant
Exam Report issued in corresponding EP Application No. 24799726.5, Jun. 17, 2025. cited by applicant

Primary Examiner: Vortman; Anatoly

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to the telecommunications and networking fields. More particularly, the present disclosure relates to a dual flow low profile coolant distribution manifold for distributing parallel coolant fluid flows to faceplate optical devices, such as pluggable optical devices, of a module or circuit pack in limited height applications.

BACKGROUND

(2) Thermal analysis has determined that it is much more efficient to distribute coolant fluid to faceplate optical devices of a module or circuit pack in a parallel flow arrangement, rather than serial flow arrangement. With Quad Small Form Factor Pluggable-Double Density (QSFP-DD) devices and the like, which are continuously getting hotter, air and fin cooling is often insufficient. Liquid or hybrid liquid/air cooling is typically more effective and allows higher power to be used.

(3) Typically, an inlet manifold and an outlet manifold are used to distribute inlet and return coolant flows to these devices. Using two separate manifolds works if there is enough board space and height. Often, however, significant height constraints are present, especially when cooling optics on a secondary side of a printed circuit board (PCB).

(4) The present background is provided as illustrative environmental context only and should not be construed to be limiting in any manner. It will be readily apparent to those of ordinary skill in the art that the principles and concepts of the present disclosure may be implemented in other environmental contexts equally.

BRIEF SUMMARY

(5) The present disclosure provides a dual flow low profile coolant distribution manifold assembly and method combining two plenums into one device to save PCB height and evenly distribute parallel coolant flow paths. This is done by machining or die casting a block in the desired shape, for example. Two holes are machined or die cast for each faceplate port. One hole feeds the port with cooler fluid and the other port allows the return of the heated fluid after it cools the associated device. A first hole and tube provides fluid inlet on one end of the device, while a second hole and tube on the opposite end of the device provides heated return fluid exit to the associated coolant distribution unit. The multiple port holes are connected to provide two separate paths—one path for cool inlet fluid and another path for heated return fluid.

(6) In one embodiment, the present disclosure provides a coolant distribution manifold assembly for use in a module or circuit pack of an optical networking system, the coolant distribution manifold assembly including: a body defining a main inlet port at one end of the body, a main outlet port at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port; where the body further defines an upper internal plenum coupled to one of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports; and where the body further defines lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports. Optionally, the body includes multiple components assembled to form a unitary body. Alternatively, the body is integrally formed. Optionally, the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port. The upper internal plenum and the lower internal plenum may each have a tapering cross-sectional area along a length of the body between the main inlet port and the main outlet port. Optionally, the body further defines or contains one or more internal lead-in features, lead-out features, flow encourager structures, flow discourager structures, or flow dividers, i.e., flow

modification structures, associated one or more of the plurality of cooling plate inlet ports or cooling plate outlet ports.

(7) In another embodiment, the present disclosure provides a liquid cooling assembly for an optical networking system, the liquid cooling assembly including: a coolant distribution unit; a plurality of input coolant lines coupled to a plurality of cooling plates; a plurality of output coolant lines coupled to the plurality of cooling plates; and a coolant distribution manifold assembly. The coolant distribution manifold assembly includes a body defining a main inlet port coupled to the coolant distribution unit at one end of the body, a main outlet port coupled to the coolant distribution unit at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port and coupled to the plurality of input coolant lines, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port and coupled to the plurality of output coolant lines; where the body further defines an upper internal plenum coupled to one of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports; and where the body further defines lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports. Optionally, the body includes multiple components assembled to form a unitary body.

Alternatively, the body is integrally formed. Optionally, the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port. The upper internal plenum and the lower internal plenum may each have a tapering cross-sectional area along a length of the body between the main inlet port and the main outlet port. Optionally, the body further defines or contains one or more internal lead-in features, lead-out features, flow encourager structures, flow discourager structures, or flow dividers, i.e., flow modification structures, associated one or more of the plurality of cooling plate inlet ports or cooling plate outlet ports. The coolant distribution manifold assembly is coupled to a printed circuit board disposed within a module or circuit pack of the optical networking system.

(8) In a further embodiment, the present disclosure provides a method for providing a coolant distribution manifold assembly for use in a module or circuit pack of an optical networking system, the method including: forming a body defining a main inlet port at one end of the body, a main outlet port at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port; where the body further defines an upper internal plenum coupled to one of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports; and where the body further defines lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports and the main outlet port and the plurality of cooling plate outlet ports. Optionally, forming the body includes assembling multiple components to form a unitary body. Alternatively, forming the body includes integrally forming the body. Optionally, the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port. The upper internal plenum and the lower internal plenum may each have a tapering cross-sectional area along a length of the body between the main inlet port and the main outlet port. Optionally, the body further defines or contains one or more internal lead-in features, lead-out features, flow encourager structures, flow discourager structures, or flow dividers, i.e., flow modification structures, associated one or more of the plurality of cooling plate inlet ports or cooling plate outlet ports. The method may further include one of: milling the upper internal plenum into a top surface of the body, milling the lower internal plenum into a bottom surface of the body, affixing a cover to the top surface of the body, and affixing a cover to the bottom surface of the body; and milling the upper internal plenum and the lower internal plenum into a side surface of the body and affixing one or more covers to the side surface of the body.

(9) It will be readily apparent to those of ordinary skill in the art that aspects and features of each of the described embodiments may be incorporated, omitted, and/or combined as desired in a given application, without limitation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is illustrated and described with reference to the various drawings, in which like reference numbers are used to denote like assembly components/method steps, as appropriate, and in which:
- (2) FIG. 1 illustrates a conventional coolant distribution manifold utilizing two plenums assembled to the PCB of a module or circuit pack, the coolant distribution manifold occupying a considerable amount of vertical space within the module or circuit pack;
- (3) FIG. 2 illustrates one embodiment of the coolant distribution manifold of the present disclosure utilizing two plenums assembled to the PCB of a module or circuit pack, the coolant distribution manifold occupying significantly less vertical space within the module or circuit pack;
- (4) FIG. 3 illustrates one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (5) FIG. 4 further illustrates one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (6) FIG. 5 further illustrates one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (7) FIG. 6 further illustrates one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (8) FIG. 7A illustrates the fluid flow associated with one embodiment of the coolant distribution manifold assembly of the present disclosure showing a plane through the inlet channel of the embodiment of the coolant distribution manifold assembly;
- (9) FIG. 7B illustrates the fluid flow associated with one embodiment of the coolant distribution manifold assembly of the present disclosure showing a plane through the outlet channel of the embodiment of the coolant distribution manifold assembly;
- (10) FIG. 8A illustrates the fluid pressure distribution associated with one embodiment of the coolant distribution manifold assembly of the present disclosure showing a plane through the inlet channel of the embodiment of the coolant distribution manifold assembly;
- (11) FIG. 8B illustrates the fluid pressure distribution associated with one embodiment of the coolant distribution manifold assembly of the present disclosure showing a plane through the outlet channel of the embodiment of the coolant distribution manifold assembly;
- (12) FIG. 9A illustrates the temperature versus port associated with one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (13) FIG. 9B illustrates the flow rate versus port associated with one embodiment of the coolant distribution manifold assembly of the present disclosure;
- (14) FIG. 10 illustrates another embodiment of the coolant distribution manifold assembly of the present disclosure;
- (15) FIG. 11 further illustrates another embodiment of the coolant distribution manifold assembly of the present disclosure;
- (16) FIG. 12 further illustrates another embodiment of the coolant distribution manifold assembly of the present disclosure;
- (17) FIG. 13 again illustrates one embodiment of the coolant distribution manifold of the present disclosure utilizing two plenums assembled to the PCB of a module or circuit pack, the coolant distribution manifold occupying significantly less vertical space within the module or circuit pack;

- (18) FIG. **14** illustrates a further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (19) FIG. **15** further illustrate a further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (20) FIG. **16** further illustrates a further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (21) FIG. **17** illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (22) FIG. **18** further illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (23) FIG. **19** further illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (24) FIG. **20** further illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (25) FIG. **21** illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (26) FIG. **22** further illustrates the fluid flow associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (27) FIG. **23** illustrates the temperature versus port associated with a still further embodiment of the coolant distribution manifold assembly of the present disclosure;
- (28) FIG. **24** illustrates a pluggable device cooling plate used in conjunction with the coolant distribution manifold assembly of the present disclosure;
- (29) FIG. **25** illustrates a coolant distribution unit used in conjunction with the coolant distribution manifold assembly of the present disclosure; and
- (30) FIG. **26** illustrates a further embodiment of the coolant distribution manifold assembly of the present disclosure, highlighting the use of quick disconnects with pluggable modules with integrated cold plates.

(31) It will be readily apparent to those of ordinary skill in the art that aspects and features of each of the illustrated embodiments may be incorporated, omitted, and/or combined as desired in a given application, without limitation.

DETAILED DESCRIPTION

(32) FIG. **1** illustrates a conventional coolant distribution manifold **10** utilizing two plenums, upper **12** and lower **14**, assembled to the PCB **16** of a module or circuit pack **18**, the coolant distribution manifold **10** occupying a consider amount of vertical space within the module or circuit pack **18**. One of the upper plenum **12** and the lower plenum **14** acts as a source plenum or path for providing cool coolant from a connector assembly **20** couplable to a coolant distribution unit associated with a shelf assembly or the like to a plurality of coolant lines **22** that feed the associated cooling plates **24** used to cool the pluggable optical modules **26** when inserted into the faceplate or other surface of the module or circuit pack **18**. The other of the upper plenum **12** and the lower plenum **14** acts as a return plenum or path for providing heated coolant from the plurality of coolant lines **22** to the connector assembly **20**. As illustrated, the upper plenum **12** and the lower plenum **14** are separate, stacked components or paths that take up a considerable amount of vertical space within the module or circuit pack **18**, vertical space which is not always available and is desirable to minimize. Further, it is desirable that the parallel coolant flows delivered to the cooling plates **24** and pluggable optical modules **26** be relatively even and that problematic pressure drops from source end to return end be avoided.

(33) FIG. **2** illustrates one embodiment of the coolant distribution manifold **110** of the present disclosure utilizing two plenums, upper **112** and lower **114**, assembled to the PCB **116** of a module or circuit pack **118** in a unitary body **130**, the coolant distribution manifold **110** thus occupying a smaller amount of vertical space within the module or circuit pack **118**. Again, one of the upper

plenum **112** and the lower plenum **114** acts as a source plenum or path for providing cool coolant from a connector assembly **120** couplable to a coolant distribution unit associated with a shelf assembly or the like to a plurality of coolant lines **122** that feed the associated cooling plates **124** used to cool the pluggable optical modules **126** when inserted into the faceplate or other surface of the module or circuit pack **118**. The other of the upper plenum **112** and the lower plenum **114** acts as a return plenum or path for providing heated coolant from the plurality of coolant lines **122** to the connector assembly **120**. As illustrated, the upper plenum **112** and the lower plenum **114** are coupled components or paths that take up considerably less vertical space within the module or circuit pack **118**. Further, the parallel coolant flows delivered to the cooling plates **124** and pluggable optical modules **126** are relatively even and problematic pressure drops from source end to return end are avoided.

(34) FIG. **3** illustrates one embodiment of the coolant distribution manifold assembly **110** of the present disclosure. The coolant distribution manifold assembly **110** includes a body **130** defining an internal upper coolant flow path **112** and an internal lower coolant flow path **114**, adapted to carry cool source coolant flow and heated return coolant flow. The body **130** includes a main inlet port/tube **132** in fluid communication with the inlet flow path (whichever is utilized) and a main outlet port/tube **134** in fluid communication with the outlet flow path (whichever is utilized). Cooling plate inlet ports/tubes **136** are provided along the length of the body **130** for feeding fresh coolant to the various coolant lines **122** in parallel. Cooling plate outlet ports/tubes **138** are provided along the length of the body **130** for receiving return coolant from the various coolant lines **122** in parallel. The body **130** and other components may be manufactured from any suitable rigid metallic or plastic material that is molded, cast, milled, and/or machined to have a desired internal and external shape, and may be formed as one component or assembled from multiple components, provided that the coolant distribution manifold assembly **130** forms a unitary structure for assembly into the module or circuit pack **118** having a controlled and minimized height. All of the ports **132,134,136,138** may include holes manufactured or machined into the body **130**, as well as tubes disposed in the holes. The body **130** may include/define any appropriate attachment points for coupling the coolant distribution manifold assembly **110** to the PCB **116**. FIG. **3** illustrates the coolant distribution manifold assembly **110** with the top removed from the body **130**, exposing the upper coolant flow path **112**, which as a tapered or variable width and/or depth (cross-sectional area) along the length of the body **130** between the inlet end and the outlet end. This variation serves to equalize and control the fluid pressure and flow associated with each of the cooling plate inlet/outlet ports/tubes **136,138**, as the fluid flows along the body **130** in each of the upper plenum **112** and the lower plenum **114**, which flows are isolated from one another. The vertical offset of the milled pockets minimizes the vertical height of the body **130**.

(35) Referring to FIG. **4**, it can be seen that the inlet plenum **112** including the cooling plate inlet ports **136** is manufactured into the upper portion of the body **130** with an increasing cross-sectional area between the inlet end **132** of the body **130** and the outlet end **134** of the body **130**, while the outlet plenum **114** including the cooling plate outlet ports **138** is manufactured into the lower portion of the body **130** with an increasing cross-sectional area between the inlet end **132** of the body **130** and the outlet end **134** of the body **130**. It will be readily apparent to those of ordinary skill in the art that other vertical placements and path area variations may be used as desired in a given application to provide a given flow/pressure/temperature profile. All fluid flows into the inlet plenum **112**, through a given cooling plate inlet port **136**, through the associated cooling plate outlet port **138**, and out of the outlet plenum **114**. As is illustrated, the walls defining the various cooling plate inlet/outlet ports **136,138** may protrude into the opposite of the inlet/outlet plenum **112,114** to a degree within the interior of the body **130** to conserve vertical space.

(36) Referring to FIG. **5**, it can be seen that the cooling plate inlet ports **136** and cooling plate outlet ports **138** may be offset and staggered along the end-to-end length of the body **130** between the main inlet port **132** and the main outlet port **134**, as well as offset vertically, again to conserve

vertical space. Each of the ports **132,134,136,138** may include a tube that makes rapid sealing connections to the various ports **132,134,136,138** easier.

(37) Referring to FIG. **6**, the milled channels of the upper and lower plenums **112,114** are enclosed with upper and lower covers **142,144** that are affixed to the body **130**, such as with an adhesive or by soldering or brazing. As alluded to above, tubes **146,148** may be sealingly inserted into the various port holes **150** to make rapid sealing connections to the various ports **132,134,136,138** easier.

(38) The tapered shape of the distribution channels is used because, without the taper, i.e. with uniform inlet and outlet channel cross-sections, there is port-to-port variation in port flow rate that is preferable to eliminate. The general idea of the taper is to control the fluid flow and pressure fields, which are non-uniform, within the region of the inlet and outlet channels, with the ideal goal of delivering a consistent flow rate to all ports. On the inlet side, this is achieved by using a larger channel cross-section where the channel flow rate is highest, graduating to a smaller channel cross-section where channel flow rate is lowest. On the outlet side, a similar strategy, or variation of this strategy, can also be used, provided that given flow rate distribution goals are achieved. A taper may vary linearly, but not necessarily so. The taper may increase/decrease monotonically, but not necessarily so. While the cross-section variation is achieved by modifying the width of the channel, it may also be achieved by modifying the height of the channel and/or the shape of the channel, and each channel need not be rectangular.

(39) FIG. **7** illustrate the fluid flow associated with one embodiment of the coolant distribution manifold assembly **110** of the present disclosure, with FIG. **7A** depicting a plane through the inlet channel and FIG. **7B** depicting a plane through the outlet channel. It can be seen that the flow is evened out port-to-port with tapered channels **112,114**.

(40) FIG. **8** illustrate the fluid pressure distribution associated with one embodiment of the coolant distribution manifold assembly **110** of the present disclosure, with FIG. **8A** depicting a plane through the inlet channel and FIG. **8B** depicting a plane through the outlet channel. The flow distribution pressure was analyzed with respect to tapered channels **112,114** that even the flow speed. The result for the geometry analyzed was a small increase in overall pressure drop across the manifold **110**. The taper therefore benefits the flow speed distribution to each port **136,138** to even out port temperatures but can possibly have the adverse effect of higher delta P across the manifold **110**.

(41) FIG. **9A** illustrates the temperature versus port associated with one embodiment of the coolant distribution manifold assembly **110** of the present disclosure.

(42) FIG. **9B** illustrates the flow rate versus port associated with one embodiment of the coolant distribution manifold assembly **110** of the present disclosure.

(43) FIGS. **10** and **11** illustrate another embodiment of the coolant distribution manifold **210** assembly of the present disclosure. The coolant distribution manifold assembly **210** again includes a body **230** defining an internal lower coolant flow path **212** and an internal upper coolant flow path **214**, adapted to carry cool source coolant flow and heated return coolant flow. Here, where flow speed equalization across ports and a tapered channel are not needed, both flow paths **212,214** are milled from one side of the body **230** to reduce setup costs. The body **230** again includes a main inlet port/tube **232** in fluid communication with the inlet flow path (whichever is utilized) and a main outlet port/tube **234** in fluid communication with the outlet flow path (whichever is utilized). Cooling plate inlet ports/tubes **236** are provided along the length of the body **230** for feeding fresh coolant to the various coolant lines **222** in parallel. Cooling plate outlet ports/tubes **238** are provided along the length of the body **230** for receiving return coolant from the various coolant lines **222** in parallel. The body **230** and other components may again be manufactured from any suitable rigid metallic or plastic material that is molded, cast, milled, and/or machined to have a desired internal and external shape, and may be formed as one component or assembled from multiple components, provided that the coolant distribution manifold assembly **230** forms a unitary

structure for assembly into the module or circuit pack **118** having a controlled and minimized height. All of the ports **232,234,236,238** may include holes manufactured or machined into the body **230**, as well as tubes disposed in the holes. The body **230** may include/define any appropriate attachment points for coupling the coolant distribution manifold assembly **210** to the PCB **216**. The vertical offset of the milled pockets minimizes the vertical height of the body **230**. Covers **250** are again provided to seal the channels **212,214**, in this case at the side of the body **230**.

(44) FIGS. **12** and **13** illustrate the coolant distribution manifold assembly **210** of the present disclosure coupled to the associated coolant lines **222** and cooling plates **224**. As illustrated, various clamp devices **252** may be used to secure the tubes used to the associated coolant lines. The cooling plates **224** are adapted to cool the pluggable modules **226** disposed through the faceplate of the module or circuit pack **218**. Optionally, an arrangement may be used whereby the manifold inlet and outlet ports **232,234** are adjacent to one another. This is of benefit in systems in which space constraints demand that the outlet port **234** and its associated tube run to be co-located with the inlet port **232** and its associated tube; e.g., inlet and outlet tubing run as a parallel pair.

(45) FIG. **14** illustrates a further embodiment of the coolant distribution manifold assembly **310** of the present disclosure that preserves good pressure and flow characteristics on both the supply **322** and return side **324** of the manifold body **330**. The dividing wall **331** between the supply side **322** and the return side **324** is contoured in such a manner that the available cross-sectional area for flow is commensurate with the local flow rate. The guiding principle is that pressure drop—which generally is important to reduce—increases non-linearly with fluid speed, and, in this embodiment, efforts are made to reduce speed particularly where it would otherwise be highest, while allowing speed to increase particularly where it would otherwise be lowest. The trade-off allows for a lower total pressure drop from manifold inlet (supply) **322** to manifold outlet (return) **324**.

(46) Referring to FIGS. **15**, the cross-section in FIG. **15a** intersects supply port **1** of a multi-port (e.g., 7-port) manifold **310**. It is apparent that the cross-section is completely allocated to supply side flow **322**, and that no cross-section is made available for return flow **324**. This is the most suitable distribution at this section because here 100% of the system flow is carried on the supply side **322**, and there is no requirement for the manifold **310** to carry return flow **324**. The cross-section in FIG. **15b** intersects return port **1**. At this position, 86% of the system flow is carried on the supply side **322**; i.e., flow for ports **2** through **7**, while 14% of the system flow is carried on the return side **324**; i.e. port **1** of **7** only. At this position, supply side and return side flow sections are interleaved such that the supply side cross-section is L-shaped, while the return side cross-section is rectangular. The dividing wall **331** can be contoured in numerous different manners, while preserving the intent that the cross-sectional area for flow is commensurate with the local supply and local return flow rates, which are generally different from one another.

(47) The cross-section in FIG. **15c** intersects supply port **2**. Here, it is apparent that the supply cross-section is decreased relative to FIG. **15a**, and some cross-sectional area is now allocated for the return flow **324**. The cross-section in FIG. **15d** intersects return port **2**. Here, it is apparent that the dividing wall **331** is in a slightly different position from FIG. **15b**, the supply cross-section is decreased relative to FIG. **15b**, and the return cross-section is increased relative to FIG. **15b**.

(48) The cross-section in FIG. **15e** intersects supply port **6**. Here, it is apparent that the trend noted above of decreasing supply cross-section continues. At this position, 29% of the system flow is carried on the supply side **322**. The cross-section in FIG. **15f** intersects return port **6**. Here, it is apparent that the trend noted above of increasing return cross-section continues. At this position, 86% of the system flow is carried on the return side **324**. It is also apparent in FIG. **15e** and FIG. **15f** that the supply cross-section is changed from L-shaped to rectangular, while the return cross-section is changed from rectangular to L-shaped.

(49) Generally, the supply and return cross-sections are interleaved in such a manner as to maximize the total supply and return flow cross-sections, while allocating the supply and return cross-sections unequally with flow rates in consideration, while also minimizing the volume of the

dividing wall **331** between them. This is accomplished by varying the shape of the dividing wall **331** in a 3-dimensional (3-D) manner. As an example of the 3-D nature of a possible dividing wall **331**, FIG. **16** shows perspective views of the dividing wall **331** proper, using a rounded modified L-shape, with all other manifold features hidden (outer wall, lids, fluid fittings, etc.).

(50) Using this technique, the pressure drop through the manifold **310** can be reduced by 50% as compared to conventional dividing wall geometries. The advantages of doing this are one or a combination of the following: lower system pressure, lower pumping power, greater total flow, greater flow per port, improved port-to-port flow uniformity, and lower pluggable optical module temperature.

(51) In several further embodiments, the manifold can be designed to both serve the purpose of controlling the flow in a manner such that the port-to-port flow differences are reduced while preserving a low hydraulic resistance (i.e., low pressure drop). The embodiments illustrated and described are example 8-port manifolds with tapered sections, using the more complex 3-D tapering implementation described above.

(52) For example, the inlet to each port may be constructed with a lead-in feature **400**. In the flow model, the feature **400** was modeled as a 1.5-mm×45° chamfer. FIG. **17** and FIG. **18** show a section through the supply ports of the manifold, and flow speed is displayed. The lead-in feature **400** serves to reduce the hydraulic resistance of the flow transition from the manifold supply channel to the port channel. Doing so has the overall benefit of reducing total manifold hydraulic resistance while also reducing the port-to-port flow differences. The lead-in feature **400** could also be done with different dimensions, or as a quarter-round, or quarter-oval, or any non-sharp profile that serves to reduce the sharpness of the entry into the tube-channel for the associated port. While the feature **400** was modeled on only one side of the entry where the benefit was appreciated to be greatest, there is nothing precluding one from applying the feature **400** to both sides if such a duplication was found to also have a benefit. While the feature **400** was modeled in a square opening for the purposes of concept-demonstration and ease-of-modeling, there is nothing precluding one from applying the feature **400** to a round opening. The chamfered feature **400**, as modeled, might be fabricated as a milled undercut. This does not preclude the application of other fabrication techniques.

(53) The outlet to each port may be constructed with a lead-out feature **406**. FIG. **21** and FIG. **22** show a section through the return ports of the manifold, and flow speed is displayed. Similar to what was noted above, the feature **406** serves to reduce the hydraulic resistance of the flow transition from the port channel to the manifold return channel. Doing so has the overall benefit of reducing total manifold hydraulic resistance while also reducing the port-to-port flow differences. The lead-out feature **406** could also be done with different dimensions. Details of the lead-in feature **400** also apply to the lead-out feature **406**.

(54) Local (per-port) features may be built into the manifold for the purpose of encouraging or discouraging flow through specific port channels. Two examples are provided. FIG. **17** and FIG. **19** show a section through the supply ports of the manifold, and flow speed is displayed. Without a local feature, which herein we call an “encourager” **402**, port **1** experiences a lower flow rate compared to other ports. The encourager **402** serves to increase the flow rate through port **1**. In fluid dynamics terms, one way to consider the phenomenon is that the dynamic pressure of the manifold's supply flow is converted locally to static pressure by the encourager **402** and this provides additional potential for flow to circulate through port **1**. In lay terms, one may think of the encourager **402** as a “scoop”. While this feature **402** slightly increases the overall hydraulic resistance of the manifold, the benefit of improving port flow where it is weakest far outweighs the very minor hydraulic resistance penalty. This feature **402** may be applied independently of whether the channels are rectangular, or tapered in one dimension, or tapered in a complex manner.

(55) FIG. **17** and FIG. **20** show a section through the supply ports of the manifold, with attention to port **8**, and flow speed is displayed. Without a local feature, which herein we call a “discourager”

404, this port **8** experiences a higher flow rate compared to other ports. While this may be good for port **8**, it might be detrimental from a system perspective because the system goal may be to cool the ports uniformly. A system goal may be to reduce or eliminate port-to-port differences so that the end user does not have to make port-specific cooling decisions. The discourager **404** serves to decrease the flow rate through port **8** by increasing the hydraulic resistance of the flow transition from the manifold to port **8**. The feature **404**, in effect, serves the opposite purpose as the lead-in feature **402** associated with FIG. **18**. The discourager **404** may be accomplished in various ways: as shown herein, as a bump, as a post, as a mesh, as any type of obstruction adjacent to the port supply or return opening. It may be fabricated by machining the manifold cavity around the feature as required, or additively; e.g., post in hole. This feature **404** may be applied independently of whether the channels are rectangular, or tapered in one dimension, or tapered in a complex manner. A secondary benefit of the discourager **404** is that its presence likely (but not necessarily) results in increased flow through the remaining ports.

(56) In the manner described herein, encouragers **402** and discouragers **404** may be applied to ports in any combination toward achieving port-to-port flow uniformity. Furthermore, the use of encouragers **402** and discouragers **404** in combination with complex tapered flow dividers (between manifold supply and return) serve toward achieving port-to-port flow uniformity while also maintaining low overall hydraulic resistance. In effect, a double goal is achieved. When higher power pluggable optical modules are used; e.g., >50 W, the associated temperature benefit can be significant.

(57) In the simulation of this embodiment (non-optimized implementation), the minimum port flow is 0.0070 L/s (port **1**) and the maximum port flow is 0.0095 L/s (port **8**). The ratio of max to min is 1.4, which is getting closer to the ideal of unity, and which represents a significant improvement over known methods on similar compact geometry. This is illustrated in FIG. **23**.

(58) FIG. **24** illustrates a pluggable device cooling plate **124,224** used in conjunction with the coolant distribution manifold assembly **110,210,310** of the present disclosure,

(59) FIG. **25** illustrates a coolant distribution unit **500** used in conjunction with the coolant distribution manifold assembly **110,210,310** of the present disclosure.

(60) FIG. **26** illustrates a further embodiment of the coolant distribution manifold assembly **610** of the present disclosure, highlighting the use of quick disconnects **650** with pluggable modules **626** with integrated cold plates **624**. Here, the pluggable module-side ports in communication with the supply and return plenums include quick disconnects **650** that are disposed at a height from the PCB **616** such that the corresponding supply and return connectors **625** of the pluggable modules **626** are at the same height when the pluggable modules **626** are inserted, such that a blind connection is possible when the pluggable modules **626** are inserted.

(61) Although the present disclosure is Illustrated and described with reference to preferred embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples may perform similar functions and/or achieve like results. All such equivalent embodiments and examples are within the spirit and scope of the present disclosure, are contemplated thereby, and are intended to be covered by the following non-limiting claims for all purposes.

Claims

1. A coolant distribution manifold assembly for use in a module or circuit pack of an optical networking system, the coolant distribution manifold assembly comprising: a body defining a main inlet port at one end of the body, a main outlet port at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port; wherein the body further defines an upper internal plenum coupled to one of the main inlet port and the

plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; wherein the body further defines a lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; and wherein one or more of: the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port; and/or at least a portion of one or more of the upper internal plenum and the lower internal plenum has an L-shaped cross-sectional area along a length of the body such that the upper internal plenum and the lower internal plenum are at least partially interleaved to minimize a height of the body and equalize a flow rate of coolant at each of the cooling plate inlet ports and each of the cooling plate outlet ports.

2. The coolant distribution manifold assembly of claim 1, wherein the body comprises multiple components assembled to form a unitary body.

3. The coolant distribution manifold assembly of claim 1, wherein the body is integrally formed.

4. The coolant distribution manifold assembly of claim 1, wherein the upper internal plenum and the lower internal plenum each have a tapering cross-sectional area along a length of the body between the main inlet port and the main outlet port.

5. The coolant distribution manifold assembly of claim 1, wherein the variable cross-sectional area of each of the upper internal plenum and the lower internal plenum along the length of the body between the main inlet port and the main outlet port equalizes a flow rate of coolant at each of the cooling plate inlet ports and each of the cooling plate outlet ports.

6. The coolant distribution manifold assembly of claim 1, wherein the body further defines or contains one or more flow modification structures associated with one or more of the plurality of cooling plate inlet ports or cooling plate outlet ports.

7. The coolant distribution manifold assembly of claim 1, wherein the plurality of cooling plate inlet ports and the plurality of cooling plate outlet ports are offset along a length of the body between the main inlet port and the main outlet port such that a height of the body is minimized.

8. A liquid cooling assembly for an optical networking system, the liquid cooling assembly comprising: a coolant distribution unit; a plurality of input coolant lines coupled to a plurality of cooling plates; a plurality of output coolant lines coupled to the plurality of cooling plates; and a coolant distribution manifold assembly, comprising: a body defining a main inlet port coupled to the coolant distribution unit at one end of the body, a main outlet port coupled to the coolant distribution unit at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port and coupled to the plurality of input coolant lines, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port and coupled to the plurality of output coolant lines; wherein the body further defines an upper internal plenum coupled to one of the main inlet port and the plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; wherein the body further defines a lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; and wherein one or more of: the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port; and/or at least a portion of one or more of the upper internal plenum and the lower internal plenum has an L-shaped cross-sectional area along a length of the body such that the upper internal plenum and the lower internal plenum are at least partially interleaved to minimize a height of the body and equalize a flow rate of coolant at each of the cooling plate inlet ports and each of the cooling plate outlet ports.

9. The liquid cooling assembly of claim 8, wherein the body is integrally formed.

10. The liquid cooling assembly of claim 8, wherein the plurality of cooling plate inlet ports and the plurality of cooling plate outlet ports are offset along a length of the body between the main inlet port and the main outlet port such that a height of the body is minimized.

11. A method for providing a coolant distribution manifold assembly for use in a module or circuit pack of an optical networking system, the method comprising: forming a body defining a main inlet port at one end of the body, a main outlet port at another end of the body, a plurality of cooling plate inlet ports disposed between the main inlet port and the main outlet port, and a plurality of cooling plate outlet ports disposed between the main inlet port and the main outlet port; wherein the body further defines an upper internal plenum coupled to one of the main inlet port and the plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; and wherein the body further defines a lower internal plenum coupled to another of the main inlet port and the plurality of cooling plate inlet ports or the main outlet port and the plurality of cooling plate outlet ports; and one of: milling the upper internal plenum into a top surface of the body, milling the lower internal plenum into a bottom surface of the body, affixing a cover to the top surface of the body, and affixing a cover to the bottom surface of the body; or milling the upper internal plenum and the lower internal plenum into a side surface of the body and affixing one or more covers to the side surface of the body.

12. The method of claim 11, wherein forming the body comprises assembling multiple components to form a unitary body.

13. The method of claim 11, wherein forming the body comprises integrally forming the body.

14. The method of claim 11, wherein the upper internal plenum and the lower internal plenum each have a variable cross-sectional area along a length of the body between the main inlet port and the main outlet port.

15. The method of claim 14, wherein the upper internal plenum and the lower internal plenum each have a tapering cross-sectional area along a length of the body between the main inlet port and the main outlet port.

16. The method of claim 11, wherein the body further defines or contains one or more flow modification structures associated with one or more of the plurality of cooling plate inlet ports or cooling plate outlet ports.
